

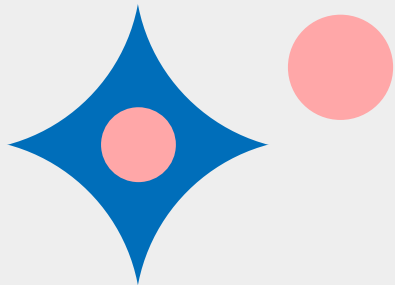
Lab - 5

Ethereum

DApp

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Why do We Need Ethereum?

Can we use Bitcoin to

- pay out a coin iff (if and only if) a user has a signing key? ☒
- implement a second currency/asset? ☒
- pay out a coin iff a candidate wins the US election? ☐
- pay out a coin iff a majority of users vote to invest in a service? ☐

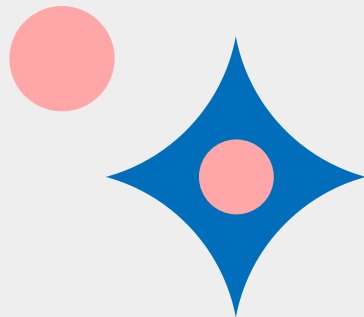


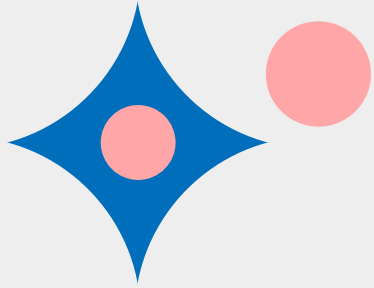
Ethereum & Smart Contracts



A computing platform on top of Blockchain.

- **EVM** (Ethereum Virtual Machine)
- **Stack**-based architecture
- Supports **Turing-complete** contract-oriented programming languages: Solidity, Vyper and LLL.





DApp

A DApp (Decentralised application) is a web service which bridges blockchain with users

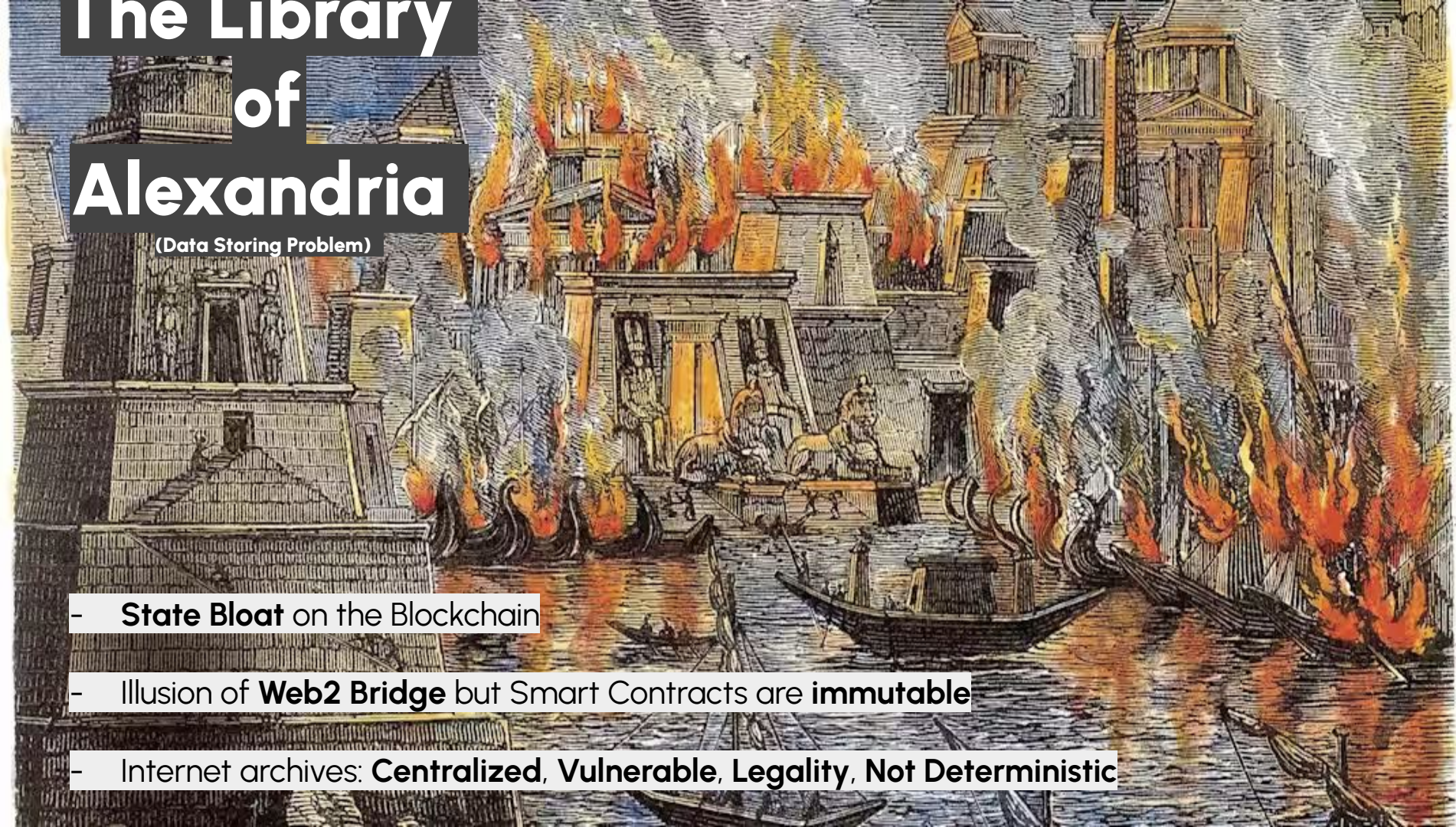
- Provides an **interface for interaction** between the blockchain and other web-services
- Mostly utilises **JavaScript (e.g. Ether.js)**

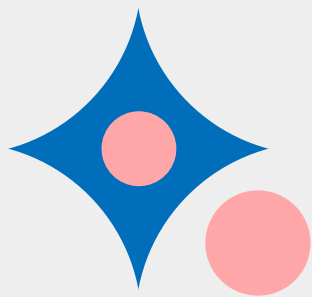


The Library of Alexandria

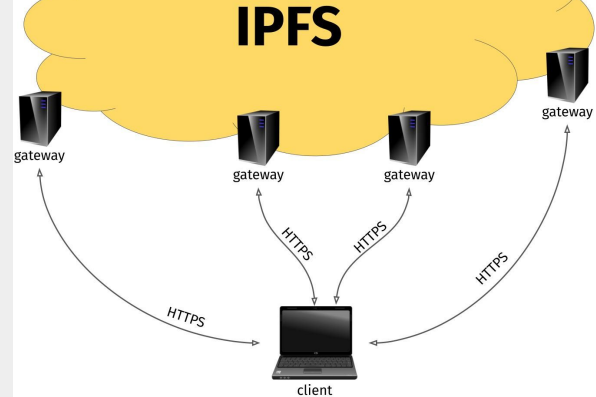
(Data Storing Problem)

- **State Bloat** on the Blockchain
- Illusion of **Web2 Bridge** but Smart Contracts are **immutable**
- Internet archives: **Centralized, Vulnerable, Legality, Not Deterministic**





IPFS (InterPlanetary File System)



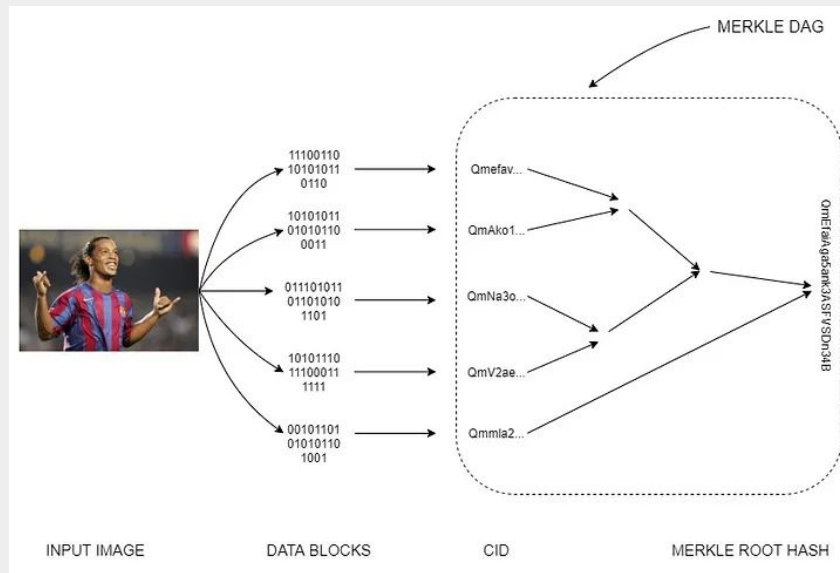
- **Peer-to-Peer Decentralized** storage system / protocol (like **BitTorrent**)
- Web2 looks for **file location** Vs
IPFS finds file via its **Cryptographic Hash**
- **Tamper-Proof**: Hashed(via **SHA-256**) files have a **unique CID** (**Content Identifier**)
- **Deduplication**: Same file uploaded more than once will generate the same hash and stored once

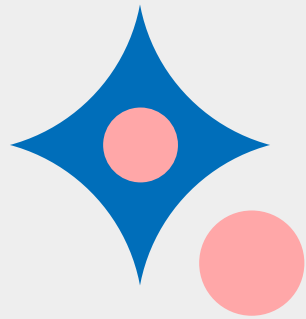




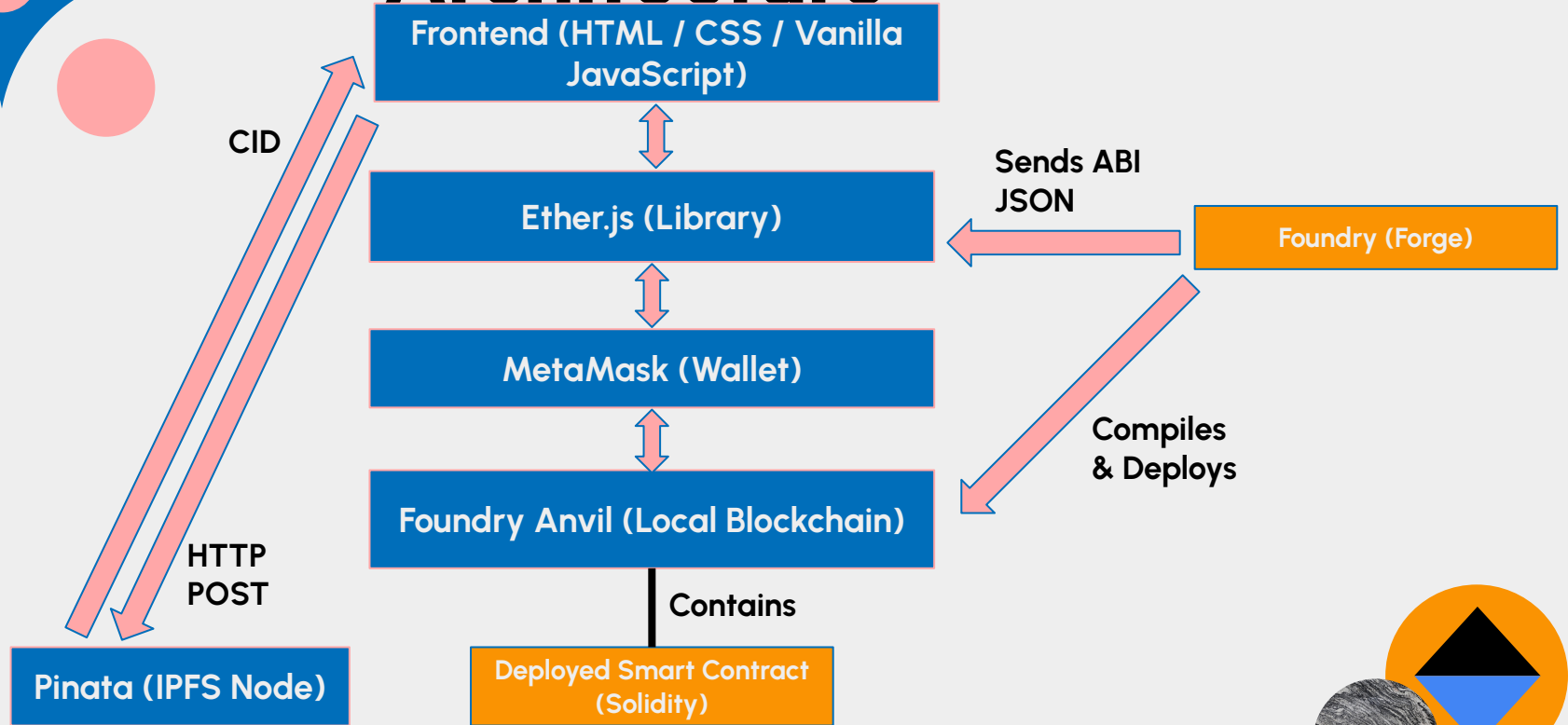
How IPFS works

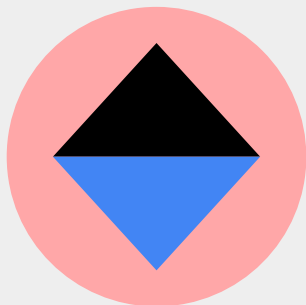
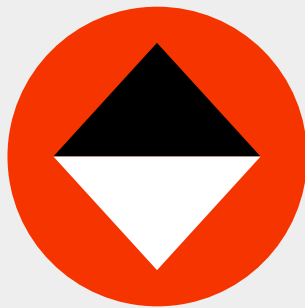
- **Chunking Data:** **256KB** per block and the **Master CID** contains all linked hashes
- For a file to survive, an **active node** must always **host** that **data**
- **Garbage Collection:** Automatically deletes **oldest, unaccessed** files (via IPFS soft.)
- **Pinning:** A mechanism to permanently keep a file (via **Pinset**)





DApp Architecture





THANK YOU!

